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Date: 5/26/2026

Course: FDN 130

Assignment 6- Views

Introduction

This is a document intended to demonstrate knowledge gained from module 6 of this course on views.

Purpose of a SQL View

A SQL view is a table that is defined by a saved SELECT statement. It provides an abstraction layer over underlying tables, showing only the data that the developer wants users to see. A developer would use a view in order to allow users to interact with data in a database without directly accessing or accidentally modifying the underlying tables.

Views are also useful for security. Sensitive data such as salaries or personal information can be excluded from a view, restricting access for users who do not need to see those data.

Differences/ Similarities Between a View, Function, and Stored Procedure

A view, a function, and a stored procedure are all ways to save and reuse SQL logic, but they differ in flexibility, complexity, and purpose.

View

A view is a table defined by a saved SELECT statement. It requires no parameters and always returns the same result based on the underlying tables. Views are the simplest of the three and are preferred when there is no customization needed with the data.

See below for how a view would be created and called upon.

```
CREATE VIEW vEmployeeNames
AS
  SELECT EmployeeFirstName + ' ' + EmployeeLastName AS EmployeeName
  FROM vEmployees;

SELECT * FROM vEmployeeNames
```

Function

A function is a reusable set of SQL instructions that accepts parameters and returns either a single value or a table. Functions are more flexible than views, and results can change based on what parameters are entered.

See below for how a function would be created and called upon.

```
CREATE FUNCTION fEmployeesByManager(@Manager NVARCHAR(100))
RETURNS TABLE
AS
RETURN
SELECT EmployeeFirstName + ' ' + EmployeeLastName AS EmployeeName
FROM vEmployees
WHERE Manager= @Manager;
```

```
SELECT* FROM fEmployeesByManager('Mickey Mouse');
```

Stored Procedure

A stored procedure is a named set of SQL statements that are executed together as a single unit. Like functions, a stored procedure can accept parameters; but it can be even more complex than a function. They can include multiple SQL statements at once and are best used for complex operations that go beyond retrieving data.

See below for how a stored procedure would be created and called upon.

```
CREATE PROCEDURE pUpdateManager
@EmployeeID INT,
@NewManager NVARCHAR(100)
AS
UPDATE Employees
SET Manager= @NewManager
WHERE EmployeeID= @EmployeeID;

EXEC pUpdateManager @EmployeeID= 1, @NewManager= 'Minnie Mouse';
```

Summary

A SQL view is a table defined by a saved SELECT statement that provides an abstraction layer over underlying tables, allowing users to interact with data without directly accessing or modifying it. Views, functions, and stored procedures are all ways to save and reuse SQL logic, but they differ in flexibility and complexity. When deciding which to use, a developer should start with the simplest tool that meets the need, using functions or stored procedures when a view's limitations require it.